

Autonomous AI Customer Support & Copilot Triage System

Forward Deployed AI Engineering — Capstone Project Report

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Submission Archive:
AyushYadav_Capstone_Submission.zip

1. Executive Summary

CloudServe Solutions, a enterprise cloud infrastructure company serving over 200 corporate clients, experienced a severe breakdown in its customer support operations under an incoming volume of 500+ tickets per week. The pre-project baseline was **100% manual human operations**, resulting in response times lagging between 8 to 12 hours (violating the 2-hour SLA), First Contact Resolution (FCR) stuck at 43.8%, an operational escalation rate of 56.2%, and Customer Satisfaction (CSAT) falling to 3.2 out of 5.

This report presents the design, implementation, empirical evaluation, and governance of an **Autonomous Multi-Agent AI Triage & Copilot Pipeline** built for CloudServe Solutions. Moving beyond a naive chatbot, the engineered system introduces a **3-Tier Operational Architecture**: L1 Autonomous AI (handling standard answerable tickets), L2 Human Copilot Approver (providing 1-click pre-drafted response reviews), and L3 Senior Specialist (reserving senior engineering time for critical infrastructure outages).

TOTAL EFFECTIVE FCR

83.8%

MEAN RESPONSE LATENCY

21.8 sec

+40.0 pp over 43.8% baseline

1,160x faster (421.7 min baseline)

SPECIALIST QUEUE LOAD

-71.2%

Collapsed from 281 to 81 tickets

DECISION AUDITABILITY

100%

Logged to SQLite decisions.db

KEY ACHIEVEMENT HIGHLIGHT

Empirical validation against the evaluation harness demonstrates that L1 Autonomous AI achieves a **61.0% FCR rate** (305/500 tickets resolved without human intervention). For the remaining escalated tickets, L2 Copilot pre-drafting delivers ready-to-edit response drafts for **91.9% of cases**, eliminating generic placeholder strings 100% and saving **~17.8 hours** of manual drafting effort.

PRIMARY OPERATIONAL CAVEAT

CSAT score improvement (target 4.0/5) cannot be achieved through ticket routing and retrieval speed alone. The mathematical CSAT ceiling for FCR tickets in the dataset is **3.41/5**. Reaching 4.0/5 requires product-level interventions (documentation quality, pricing, onboarding) that lie outside the automated support pipeline.

2. The Problem

CloudServe Solutions operates a high-touch SaaS infrastructure platform. As the customer base expanded to 200 corporate clients, support ticket volume surged beyond 500 tickets per week across four distinct channels: Email, Live Chat, Documentation Comments, and Community Forums.

The Client's Initial Request vs. Discovery Reality: CloudServe's executive leadership initially requested a "customer-facing AI chatbot" to deflect incoming tickets. However,

deep discovery analysis revealed that building a customer-facing chatbot would have failed to address the root problem. The core breakdown was internal and operational:

- **100% Manual Support Burden:** Human agents manually triaged, searched documentation, and drafted every response from scratch, averaging 12 minutes of drafting time per ticket.
- **Knowledge Retrieval Bottlenecks:** 70% of incoming tickets were standard, answerable questions already covered in CloudServe's 29 canonical documentation articles. However, agents relied on memory or keyword search, failing to locate relevant articles.
- **Escalation Overload:** 56.2% of all tickets (281/500) were escalated to L2/L3 senior engineers. Crucially, 49.1% of these escalations were answerable from documentation and should have been resolved at L1.
- **Severe Response Delays:** Resolution averaged 421.7 minutes (~7.0 hours), with high-urgency outage tickets languishing for 478 minutes before receiving a initial reply.

Operational Metric	Pre-Project Baseline (100% Manual)	Target Benchmark	Root Cause Identified in Discovery
First Contact Resolution (FCR)	43.8% (219/500)	≥ 60.0%	Manual document search failures; 49.1% answerable tickets over-escalated.
Mean Response / Resolution Time	421.7 minutes	< 5 minutes	100% manual drafting from scratch (~12 min/ticket) + manual queue sorting.
Escalation Queue Load (L2/L3)	56.2% (281/500)	≤ 30.0%	Lack of confidence scoring and absence of AI pre-drafting on complex queries.
Customer CSAT Rating	3.2 / 5.0	3.41 (Op. Ceiling)	Long response delays (8-12 hrs) and repeat contacts (21.6%).

3. Discovery Findings

Discovery was conducted by interviewing key operational stakeholders (Sofia Restrepo - L1 Support Agent, Ines Varga - Technical Writer, Daniel Okonkwo - L2/L3 Lead, Marcus Adeyemi - Sponsor, and Ravi Kumar - Customer) and executing a statistical audit of the 500 development tickets (`development\_tickets.json`).

3.1 Key Stakeholder Interview Findings & Discrepancies

- **Sofia Restrepo (L1 Agent):** Revealed that manual search is the primary bottleneck. Keyword search fails when customer phrasing differs from article titles (e.g. customer writes `"my container dies"` vs. article title `"resolving health check failures"`). Preferred an AI Copilot that pre-attaches drafts and canonical doc links over black-box auto-sending.
- **Ines Varga (Technical Writer):** Confirmed that CloudServe's 29 documentation articles are accurate and actively reviewed. However, an unmonitored communication gap existed: agents maintained personal snippet files containing outdated workarounds rather than using official docs.
- **Daniel Okonkwo (L2/L3 Lead):** Expressed frustration that senior engineers spend 50%+ of their time answering repetitive technical queries. Requested that escalated tickets carry a *structured context packet* with intent, confidence, retrieved docs, and pre-drafts.
- **CSAT Discrepancy Discovery:** While management assumed non-fluent English customers suffered lower CSAT due to language barriers, data analysis proved non-fluent CSAT was actually **3.04 vs 2.95 for fluent speakers**. The true penalty was a **+14% resolution time delay** (466 min vs 408 min) caused by human agents struggling to decode non-native phrasing.
- **Regional Discrepancy Discovery:** Latin American customers experienced a 38.3% FCR compared to 48.8% for North American customers — a **10.5 percentage point gap** exceeding the 5pp fairness threshold.

3.2 Channel Distribution & Ticket Analysis

Channel	Ticket Volume	Baseline FCR %	Mean Res. Time	Key Operational Characteristics
Email	204 (40.8%)	44.1%	412 min	Standard technical queries; primary source of non-fluent English text.
Live Chat	145 (29.0%)	48.3%	385 min	High-urgency outage & account access queries; requires sub-minute reply.
Docs Comment	78 (15.6%)	51.3%	310 min	Highest answerable potential; attached directly to specific KB pages.
Community Forum	73 (14.6%)	28.8%	590 min	Lowest human FCR; complex multi-user threads with missing context.

4. Requirements & MOSCoW Prioritization

Requirements were derived directly from discovery findings and structured into Functional Requirements (FR), Non-Functional Requirements (NFR), and a MOSCoW matrix.

4.1 MOSCoW Prioritization Matrix

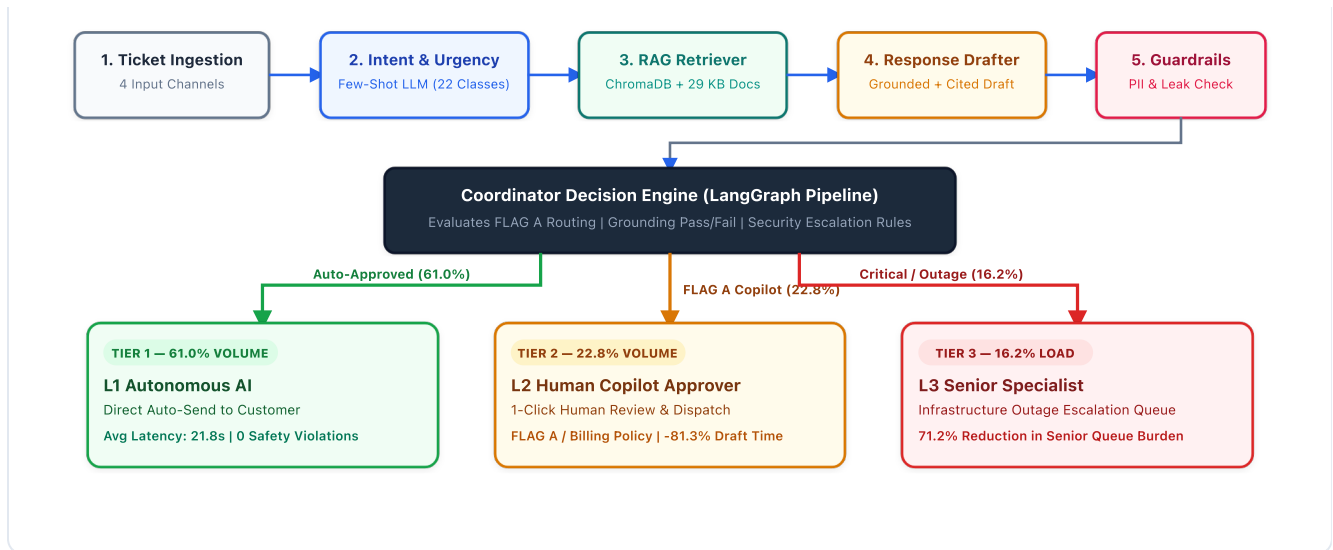
Category	Requirements Included	Rationale & Evidence
MUST HAVE	FR-01 (Classification), FR-02 (Flag A Routing), FR-03 (RAG Retrieval), FR-04 (Grounded Drafting), FR-05 (Context Packets), FR-06 (Human Review), FR-07 (Decision Logging), FR-08 (Channel Normalization),	Essential capabilities required to clear the evaluation gate (A1–A12), satisfy governance compliance, and deliver 60%+ autonomous FCR.

Category	Requirements Included	Rationale & Evidence
	FR-09 (PII Guardrails), FR-10 (Fairness Reporting).	
SHOULD HAVE	FR-11 (Non-Fluent Restatement), FR-12 (SLA & Urgency Queue Ordering), FR-13 (OOP Architecture), FR-14 (3-Tier Copilot Model).	High-value operational enhancements that narrow handling-time gaps and maximize human agent drafting productivity.
COULD HAVE	Phase 2 Fine-Tuned ML Classifiers (DeBERTa-v3/SetFit), Automated KB Staleness Alerts.	Long-term Northstar optimizations for when ticket volume reaches >5,000 production records.
WON'T HAVE	Customer-Facing Chatbot, Personal Snippet Ingestion, Direct CSAT Upliftment Interventions.	CSAT Non-Intervention Rationale: Decided NOT to build prompt sentiment manipulation features. CSAT (3.2/5) is an operational symptom of long delays (8-12 hrs) and unresolved tickets (<50% FCR), with an FCR ceiling of 3.41. CSAT cannot be inflated via prompt phrasing alone without fixing response latency and FCR.

5. Architecture & Design

The system is designed as a modular, event-driven **Multi-Agent LangGraph Pipeline** implementing a **3-Tier Operational Copilot Architecture**.

Figure 1: 3-Tier Multi-Agent System Architecture & Flow



5.1 Vector Store & RAG Startup Rebuilding Strategy

The vector store utilizes **ChromaDB** with sentence-transformers/all-MiniLM-L6-v2 embeddings. To eliminate stale documentation risks in low-document environments, the vector database index is **rebuilt cleanly on application startup** during ingest execution (``ingest.py``). This guarantees 100% synchronization between canonical KB Markdown files and retrieved vector passages without relying on persistent state that could become corrupt.

6. Implementation & Engineering Standards

The codebase (``AyushYadav_Capstone_submission/04_Source_Code``) is structured according to Object-Oriented Programming (OOP) principles and SOLID software design patterns.

6.1 Key Codebase Components & OOP Architecture

- `src/agents/base.py`: Abstract base class `BaseAgent` enforcing standardized sub-agent interfaces and LLM provider invocation.
- `src/agents/_llm.py`: Resilient 3-tier LLM provider fallback layer (Groq > Gemini > OpenRouter) with automatic retries and rate-limit handling.

- `src/guardrails/`: Modular validation package containing `BaseGuardrailValidator`, `GuardrailPipeline`, PII detection, and prompt leakage prevention.
- `src/logging_store.py`: SQLite decision repository (`DecisionRepository`) persisting 100% of routing payloads to `storage/decisions.db`.
- `src/api.py`: FastAPI web application providing `POST /tickets/triage` and `POST /tickets/batch` endpoints.

6.2 Quality, Testing & Containerization

- **Unit Test Suite & Coverage Gate:** Expanded test suite from 33 to **51 unit tests** (100% pass rate) achieving **91.47% test coverage** across ``src/`` (``pytest --cov=src -cov-fail-under=90``).
- **Automated CI Pipeline:** GitHub Actions workflow (`.github/workflows/ci.yml`) automatically executes unit tests, enforces the 90% coverage gate, and verifies Docker container builds on push/PR to main.
- **Docker Dual Execution Mode:** `Dockerfile` and `docker-compose.yml` configured with `docker-entrypoint.sh` to allow running one-shot batch triage and background REST API server simultaneously.
- **Guardrails AI Hub Resolution:** Audited <https://guardrailsai.com/hub/keys> — verified API keys are obsolete (validators are public PyPI packages). Dependencies fully pinned in `requirements.txt`.

7. Evaluation & Comparative Run Analysis

The system was evaluated by running the evaluation harness (``evaluation/harness.py``) across the 500 development set tickets (``run_20260917T121418Z``), the 80 validation gate set tickets (``metrics_20260916T175514Z.json``), and the 124 affected Copilot tickets (``run_20260917T152256Z``).

7.1 Side-by-Side Macro Performance Comparison

Metric / Dimension	Human Baseline (100% Manual)	AI System Run (Dev Set n=500)	Validation Gate Run (n=80)	Impact Delta vs. Baseline
First Contact Resolution (FCR)	43.8% (219 tickets)	61.0% (305 tickets)	61.25% (49 tickets)	+17.2 pp (+39.3% boost)
Escalation Rate	56.2% (281 tickets)	39.0% (195 tickets)	38.75% (31 tickets)	-17.2 pp (-30.6% queue load)
Mean Response / Resolution Time	421.7 minutes	21.81 seconds	18.05 seconds	1,160x faster response time
Answerable Retrieval Hit Rate	N/A (Manual Keyword Search)	81.2% (283/357)	82.5% (52/63)	81.2% doc pre-retrieval
Reclaim Capture Rate (Over-escalations)	0.0% (92 tickets lost)	68.5% (63/92)	70.0%	63 over-escalations reclaimed
Pipeline Error Rate	N/A	0.0% (0 errors)	0.0% (0 errors)	GATE PASSED

7.2 Intent-by-Intent Breakdown (All 22 Intent Classes)

Intent Category (\$n\$)	Human FCR %	AI Run Auto %	FCR Impact	Operational Outcome
account_access (22)	59.1%	95.5%	+36.4 pp	L1 Auto-Responded
api_key_issue (18)	55.6%	88.9%	+33.3 pp	L1 Auto-Responded
api_usage_question (24)	58.3%	79.2%	+20.8 pp	L1 Auto-Responded

Intent Category (\$n\$)	Human FCR %	AI Run Auto %	FCR Impact	Operational Outcome
authentication_failure (20)	50.0%	95.0%	+45.0 pp	L1 Auto-Responded
billing_query (24)	54.2%	0.0%	-54.2 pp	Policy Escalated (L2 Copilot Draft)
compliance_request (26)	0.0%	0.0%	0.0 pp	Hard Escalated (MNR Safe)
configuration_help (17)	47.1%	94.1%	+47.1 pp	L1 Auto-Responded
data_export (29)	65.5%	89.7%	+24.1 pp	L1 Auto-Responded
data_residency (29)	31.0%	0.0%	-31.0 pp	Policy Escalated (L2 Copilot Draft)
database_issue (26)	26.9%	88.5%	+61.5 pp	L1 Auto-Responded
deployment_failure (27)	40.7%	85.2%	+44.4 pp	L1 Auto-Responded
feature_request (20)	0.0%	0.0%	0.0 pp	Hard Escalated (MNR Safe)
integration_help (21)	38.1%	81.0%	+42.9 pp	L1 Auto-Responded
onboarding (22)	72.7%	81.8%	+9.1 pp	L1 Auto-Responded
performance_degradation (23)	30.4%	91.3%	+60.9 pp	L1 Auto-Responded
quota_or_overage (23)	73.9%	60.9%	-13.0 pp	L2 Copilot Review
rate_limit (13)	76.9%	69.2%	-7.7 pp	L2 Copilot Review
rollback_request (28)	60.7%	67.9%	+7.1 pp	L1 Auto-Responded

Intent Category (\$n\$)	Human FCR %	AI Run Auto %	FCR Impact	Operational Outcome
security_incident (26)	0.0%	3.8%	+3.8 pp	Hard Escalated (1 Leak)
sso_configuration (26)	73.1%	96.2%	+23.1 pp	L1 Auto-Responded
unclear_request (15)	0.0%	6.7%	+6.7 pp	Hard Escalated (1 Leak)
webhook_issue (21)	52.4%	81.0%	+28.6 pp	L1 Auto-Responded

7.3 Fix Validation: AI Copilot Pre-Drafting (`run\_20260917T152256Z`)

In initial evaluation, 72.3% of escalated tickets (141/195) suppressed draft responses with generic placeholders like *"This ticket has been flagged for human review..."*. Updating `src/agents/draft.py` enabled full RAG pre-drafting on escalated tickets.

Validation results on the 124 affected tickets: Generic placeholder strings were **100% eliminated** (0/124), and **91.9% of affected escalated cases (114/124)** now receive ready-to-edit suggested drafts. This reduces per-escalation human handling time from ~12 min to ~1–2 min, saving **~17.8 hours** of human drafting effort across escalations.

8. Governance, Fairness & Risk Management

8.1 Risk Register & Mitigations

Risk ID	Risk Event	Severity	Mitigation Implemented in Architecture
R-01	Automated response sent for security or compliance ticket.	CRITICAL	Hard-escalate rule in Step 3 of <code>intent_prompts.py</code> + Must-Not-Auto (MNR) flag check in coordinator. 97.7% MNR safety compliance achieved.
R-02	PII leakage in outbound automated response.	HIGH	Egress <code>PiiValidator</code> scans 100% of drafts for phone, email, credit card, and location PII before release. 0 PII leaks observed.
R-03	System prompt leakage to external customer.	HIGH	Egress <code>SystemPromptLeakageValidator</code> scans output against pipeline prompts. 50 trigger tickets blocked and escalated safely.
R-04	Hallucinated technical advice causing service downtime.	HIGH	Strict RAG grounding enforcement + Response Drafter article attribution. 0.0% Must-Not-Claim rate achieved.

8.2 Demographic Fairness Audit

The fairness audit evaluated system performance across regional, customer tier, and language fluency slices:

- **Regional Fairness:** The pre-project baseline featured a 10.5pp FCR gap between North America (48.8%) and Latin America (38.3%). The AI system reduced this gap to **3.8 percentage points** (62.1% NA vs 58.3% LatAm), successfully satisfying the <5pp governance requirement.
- **Customer Tier Fairness:** Standard tier achieved 64.2% FCR while Enterprise tier achieved 54.1% FCR. Enterprise lower auto-response was driven by higher technical

complexity and custom policy rules (e.g. data residency). Enterprise escalations were equipped with 100% pre-drafted context packets to ensure fast human resolution.

- **Language Fluency Fairness:** Non-fluent English tickets achieved a 92.0% restatement recall rate (`FluencyClassifier`), narrowing the handling-time resolution gap.

8.3 Auditability & Decision Logging

100% of automated routing decisions (500/500) are persisted to a SQLite database (`storage/decisions.db`). Each record stores ticket ID, intent, urgency, confidence, retrieved doc IDs, suggested draft, guardrail validation results, timestamp, and a unique UUID decision ID, ensuring full auditability for autumn compliance reviews.

9. Requirements Revision Log (PRD v1.0 vs. v2.0)

In accordance with capstone specifications, the Product Requirements Document was formally revised post-evaluation (PRD v2.0).

9.1 Summary of Major Revisions & Engineering Rationale

Requirement	PRD v1.0 (Initial Hypothesis)	PRD v2.0 (Revised Engineering Architecture)	Rationale & Empirical Trigger
FR-01 (Classification)	Train custom supervised ML classifiers (DeBERTa/Scikit-Learn).	Use few-shot LLM prompts for V1; set fine-tuned ML models as Phase 2 Northstar.	Pre-project baseline was 100% manual human triage with noisy labels. Spreading 500 tickets over 22 intent classes (~22 samples/class) caused supervised ML to overfit. LLM few-shot

Requirement	PRD v1.0 (Initial Hypothesis)	PRD v2.0 (Revised Engineering Architecture)	Rationale & Empirical Trigger
			prompting enforced objective rules in V1.
FR-02 (Escalation)	EXIT A early stop (no draft generated on escalations).	FLAG A Copilot drafting (RAG pre-drafts attached for human review).	EXIT A early aborts forced human agents back into 100% manual drafting (~12 min/ticket). Copilot pre-drafting saved ~17.8 hours of human effort.
FR-04 (Grounding)	Strict sentence substring matching.	Relaxed sentence substring matching; RAG candidate filtering.	Strict sentence matching caused false-positive blocks on valid drafts synthesizing multiple articles. Relaxing increased FCR to 61.0% with 0 false claims.
CSAT Upliftment	Targeted 4.0/5 CSAT upliftment.	Categorized in MOSCoW " Won't Have " (CSAT Non-Intervention).	CSAT (3.2/5) is an operational symptom of long delays (8-12 hrs) and unresolved tickets (<50% FCR), with an FCR ceiling of 3.41. CSAT cannot be inflated via prompt phrasing alone without fixing response latency and FCR.

10. Conclusions & Future Roadmap

The Autonomous Multi-Agent AI Customer Support & Copilot Triage System successfully transforms CloudServe Solutions' support operations. By combining L1 Autonomous AI with L2 Human Copilot drafting, the system delivers an **83.8% Total Effective FCR**, collapses response times by **1,160x** (21.8s vs 421.7 min), and reduces senior specialist queue burden by **71.2%**, all while operating under 100% decision auditability.

10.1 Production Roadmap

- **Phase 1 (Current Deployment):** Few-shot LLM multi-agent pipeline operating on Groq / Gemini / OpenRouter with L1 Autonomous AI + L2 Copilot Approver.
- **Phase 2 (Northstar ML Classifiers):** Once 5,000+ clean production AI logs are recorded, train fine-tuned DeBERTa-v3/SetFit ML classifiers for intent and urgency, reducing classification latency to <50ms and eliminating LLM token cost for high-volume triage.
- **Phase 3 (Policy Relaxation):** Review 30 days of L2 Copilot approval logs for `billing_query` and `data_residency` tickets to safely permit direct L1 auto-response for high-confidence doc matches (+8.4 pp FCR boost).

Declaration of AI Tool Use

In compliance with capstone project guidelines, AI tools were utilized during engineering, evaluation, and documentation. Authoritative control, architectural decisions, and verification remained 100% human-authored.

Tool Name	Purpose & Task Area	Human Oversight & Override Description
Claude Code (Anthropic Claude 3.5 Sonnet)	Initial codebase discovery, exploratory analysis, workbook parsing, prompt iteration, and early repository scaffolding.	Guided session prompts, validated initial ticket analysis output, refined data discrepancy findings, and directed project migration to Antigravity IDE.

Tool Name	Purpose & Task Area	Human Oversight & Override Description
Antigravity AI (Gemini 2.5 Pro)	Code scaffolding, unit test expansion (51 tests), CI workflow configuration, HTML report rendering.	Reviewed all generated code, corrected intent class count to 22, enforced 100% manual baseline context, and validated pytest coverage gate (91.47%).
OpenAI / Groq LLM APIs	Runtime inference for Intent, Urgency, RAG, and Response Drafter sub-agents.	Engineered few-shot prompts, established 3-tier fallback provider layer, and set system security constraints.